INCH-POUND

MIL-C-55302/175A
26 DEC 1989
SUPERSEDING
MIL-C-55302/175
15 April 1987

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES· 2-ROW PIN ASSEMBLIES, RIGHT ANGLE, 20 THROUGH 150 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation. MIL-C-55302.

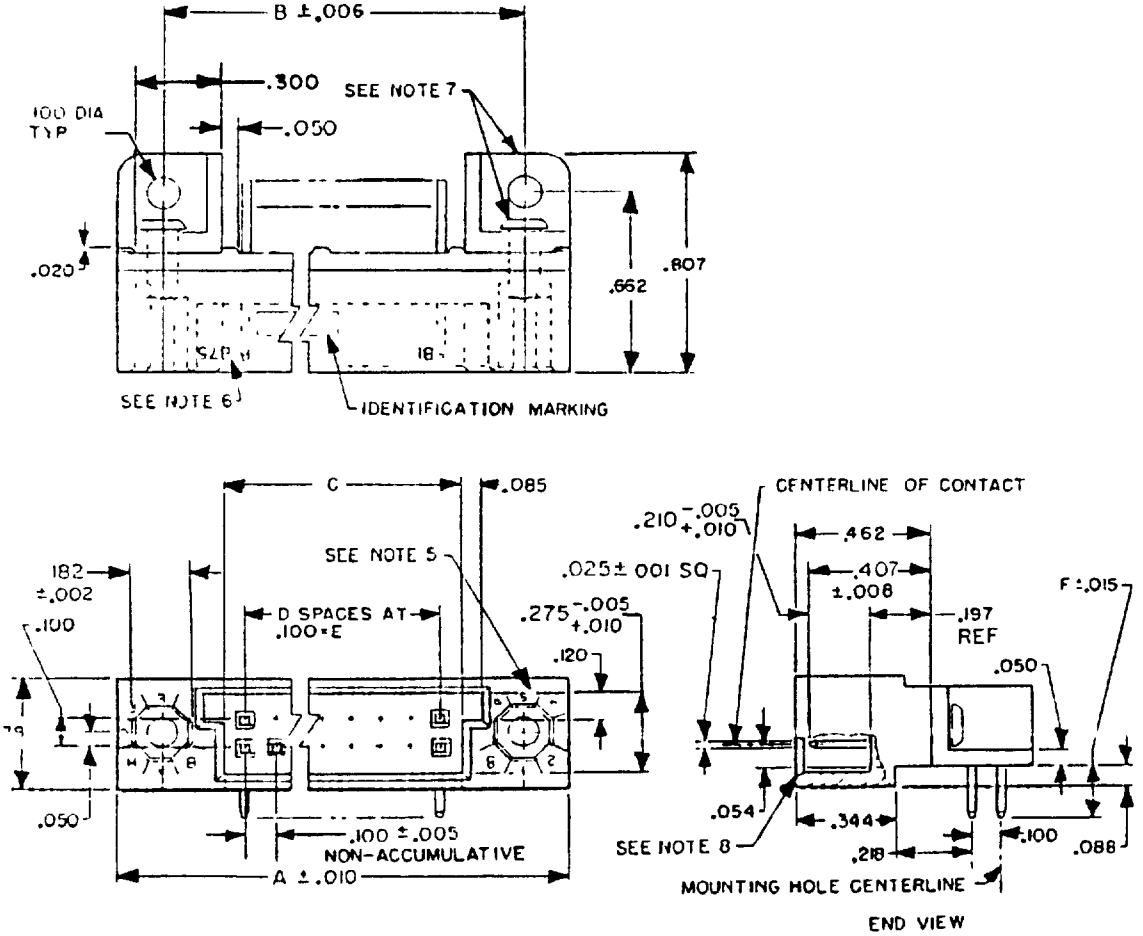
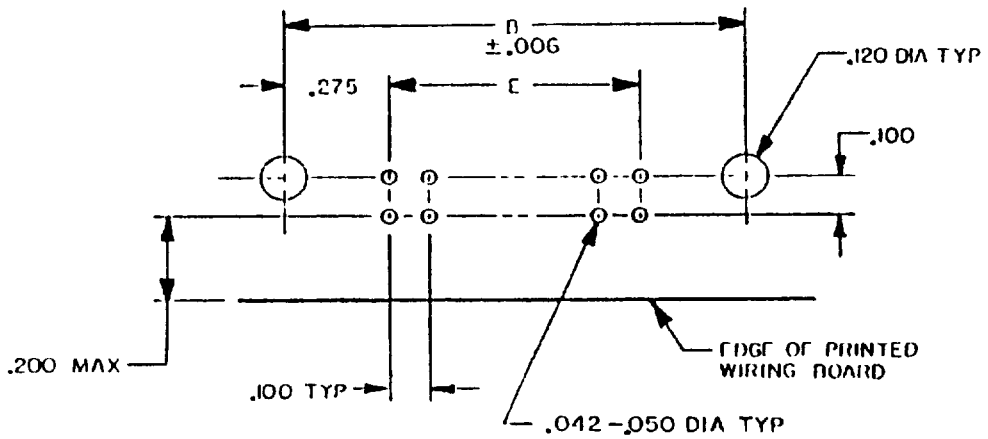


FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 20 through 150 contacts.

AMSC N/A FSC 5935

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Recommended circuit layout

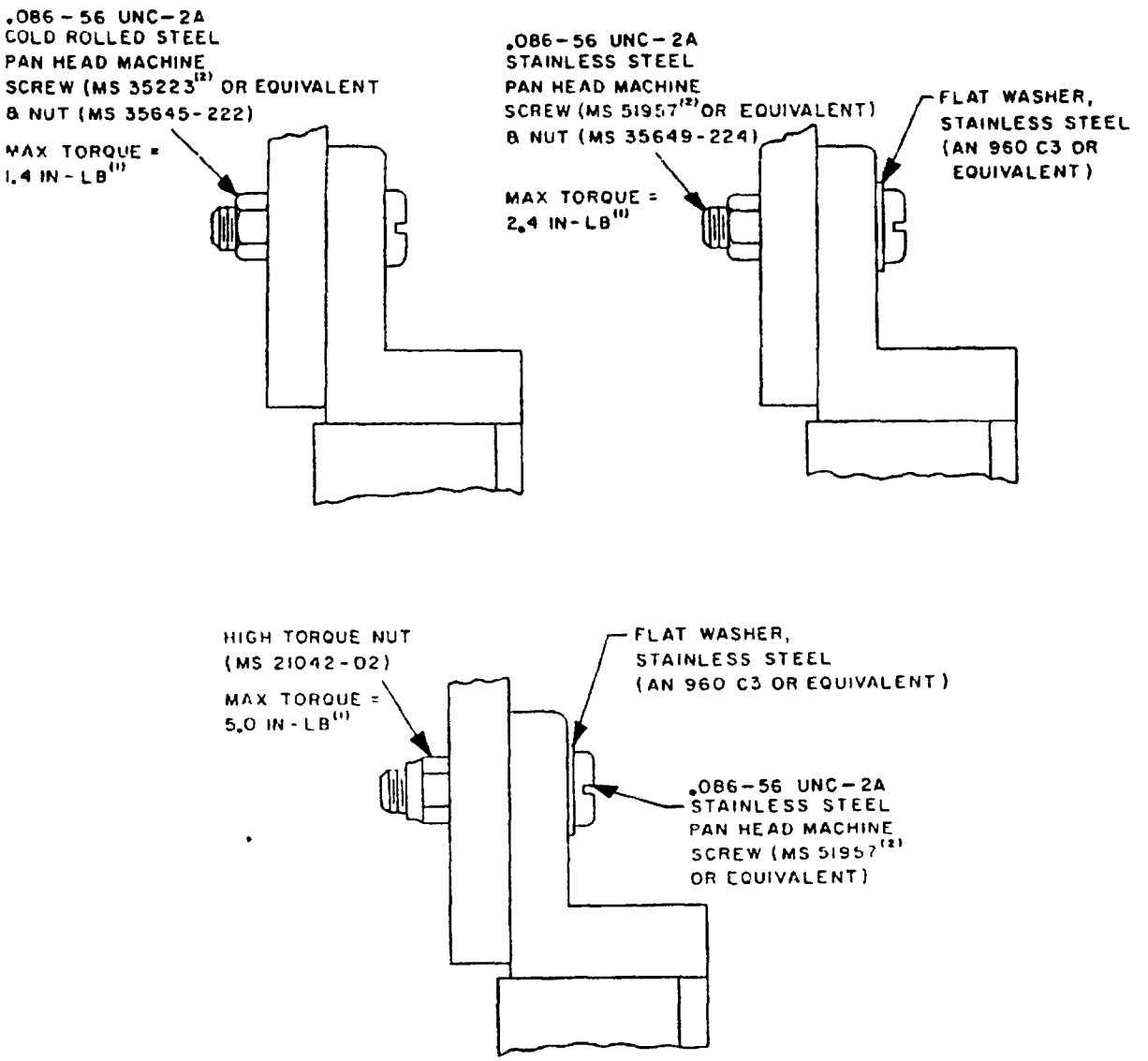
F ±.015 (0.38 mm)		Type	Type of termination
Inches	mm		
.180	4.57	I	Solder post (see note 9)
.120	3.05	IV	

Inches	mm	Inches	mm
.001	0.03	.100	2.54
.002	0.05	.120	3.05
.005	0.13	.182	4.62
.006	0.15	.197	5.00
.008	0.20	.200	5.08
.010	0.25	.210	5.33
.015	0.38	.218	5.54
.020	0.51	.275	6.98
.025	0.64	.300	7.62
.042	1.07	.344	8.74
.050	1.27	.38	9.7
.054	1.37	.407	10.34
.085	2.16	.462	11.73
.086	2.18	.662	16.81
.088	2.24	.807	20.50

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/174 and /183.
5. Location indicators embossed on surface
6. Numbers indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface
7. Mounting ears and hardware are purchased separately. See MIL-C-55302/182 for mounting hardware.
8. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ±5°.
9. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1 Connectors, pin assemblies .100 (2.54 mm) spacing, 20 through 150 contacts - Continued.

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NOTES.
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 20 through 150 contacts - Continued.

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TABLE I. Dash number, number of contacts, contact type, and dimensions. 1/

Dash no.	No. of contacts	Contact type	A	B	C	D	E	Contact ident. no.
01 04	20	I IV	1.750 (44.45)	1.450 (36.83)	1.050 (26.67)	9	.900 (22.86)	1-10
05 08	30	I IV	2.250 (57.15)	1.950 (49.53)	1.550 (39.37)	14	1.400 (35.56)	1-15
09 12	40	I IV	2.750 (69.85)	2.450 (62.23)	2.050 (52.07)	19	1.900 (48.26)	1-20
13 16	50	I IV	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)	1-25
17 20	56	I IV	3.550 (90.17)	3.250 (82.55)	2.850 (72.39)	27	2.700 (68.58)	1-28
21 24	60	I IV	3.750 (95.25)	3.450 (87.63)	3.050 (77.47)	29	2.900 (73.66)	1-30
29 32	80	I IV	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)	1-40
37 40	100	I IV	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)	1-50
57 60	134	I IV	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)	1-67
65 68	150	I IV	8.250 (209.55)	7.950 (201.93)	7.550 (191.77)	74	7.400 (187.96)	1-75

1/ Deleted dash numbers in the series 01 through 68 are canceled with no replacement.

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TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area shall have a minimum plating in accordance with MIL-C-55032. Solder post area (contact type I, .195 minimum length; type IV, .135 minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.

1/ Type C finish is deleted and superseded by type A or B.

REQUIREMENTS:

Design and construction:

Dimensions and configuration See figure 1 and table I.

Material. In accordance with MIL-C-55302.

Plating See table II.

Current rating 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31).

Two keys, Part or Identifying Number (PIN) MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Mating and unmating The maximum mating force in pounds shall be the number of contacts multiplied by .125, the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet. 200 volts rms.

2 milliamperes maximum leakage current.

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Part or Identifying Number (PIN) M55302/175 (and dash number from table I and type from table II)

Example M55302/175-01A.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians
Army - CK
Navy - EC
Air Force - 17

Review activities
Army - AR
Navy - AS, SH
Air Force - 11, 85, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent
DLA - ES

(Project 5935-3715-03)

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